

Mock Midterm

CS 101 (L5) Algorithmic Problem Solving Fall 2023

Exam Instructions

Write your name and HU ID on the answer sheet provided separately.

Exam Duration: 2 hours and 30 minutes

Number of Questions: This exam contains 4 coding questions and 1 debugging question.

Technical Requirements: You are not allowed to use code editors, compilers or online resources during this exam.

Communication: During the exam, do not communicate with anyone except the instructor or exam proctors if you have questions about the exam.

Honesty: Academic honesty is expected. Do not engage in any form of cheating, plagiarism, or unauthorized collaboration.

Answer Format: Write your answers clearly on the answer sheet.

Time Management: Monitor the time carefully and pace yourself. Do not spend too much time on any one question or section.

Exam Completion: Before submitting, take a moment to review your answers to ensure you have completed all questions and that your responses are clear and accurate.

On your answer sheet, note down each question number. For each question numbered 1 through 4, (a) write the output for each exercise, (b) identify the problem, (c) write the pseudocode, and (d) dry run it using two of the inputs provided in the Exercise section.

1. Pugam Pugai

Allah Rakha, Bi Haleema, and Chacha Cricket make everyday decisions with the aid of the traditional game *pugam pugai*. To play the game, all players must throw their hands simultaneously, either palm up or palm down. The player with the unique hand wins, that is, the player whose hand is different

from the other two players wins. If all players have their hands palm up or down, the game is considered drawn.

Given the thrown hands, can you figure out who wins?

Constraints

- Each of A , B , and C is a string with a value or either “UP” or “DOWN”

Interactions

The input comprises a single line containing three words, each of which is either “UP” or “DOWN”, denoting the values of A , B , and C respectively.

The output must contain a single line stating either the name of the winner, one of “Allah Rakha”, “Bi Haleema” or “Chacha Cricket”, or the word “Drawn” if no one wins.

Sample

Input	Output
DOWN UP UP	Allah Rakha
DOWN UP DOWN	Bi Haleema
UP UP DOWN	Chacha Cricket
UP UP UP	Drawn

In the first case, Allah Rakha has thrown his hand palm down, while both Bi Haleema and Chacha Cricket have thrown their hands palm up. Allah Rakha wins.

In the second case, Bi Haleema has thrown her hand palm up, while both Allah Rakha and Chacha Cricket have thrown their hands palm down. Bi Haleema wins.

In the third case, Chacha Cricket has thrown his hand palm down, while both Allah Rakha and Bi Haleema have thrown their hands palm up. Chacha Cricket wins.

In the fourth case, all three have thrown their hands palm up. The game is drawn.

Exercise

	Input
(i)	UP DOWN UP
(ii)	DOWN DOWN DOWN
(iii)	DOWN DOWN UP

2. Game of Coins!

Nitin and Sobhagya were playing a game with coins. If Sobhagya has more coins then he is winning, otherwise Nitin is winning. Note that this means if both Nitin and Sobhagya have the same number of coins, then Nitin is winning.

Initially Nitin has A coins while Sobhagya has B coins. Then Ritik came and gave his C coins to the player who is not winning currently, after which Satyarth came and repeated the same process (gave his D coins to the player who is not winning currently).

Write a function *to winner* which takes in the parameters A , B , C and D and returns the first letter of the person who wins the game (either “S” or “N”).

Make sure to call your function and to present the output thus:

```
print(functionname(parameters))
```

Constraints

- $1 \leq A, B, C$ and $D \leq 100000$

Interactions

The input comprises four space separated integers denoting the values of A , B , C and D .

The output must contain a single character “S” or “N” depending on who won the game.

Sample

Input	Output
2 3 4 5	S
3 3 3 3	N

In the first case, $(A, B, C, D) = (2, 3, 4, 5)$, Initially, Nitin has 2 coins and Sobhagaya has 3 coins, so S is winning, Ritik comes and gives his 4 coins to Nitin and Nitin has 6 coins now, then Satyarth gives his 5 coins to Sobhagaya, since Sobhagaya has 8 coins now he won.

In the second case, $(A, B, C, D) = (3, 3, 3, 3)$, Initially, Nitin and Sobhagaya both have 3 coins each, Ritik gives 3 coins to Sobhagaya so he has 6 coins now and then Satyarth gives his 3 coins to Nitin, finally Nitin and Sobhagaya have 6 coins each so Nitin is the final winner.

Exercise

	Input
(i)	2 4 7 1
(ii)	3 9 8 3
(iii)	1 7 8 2

3. Amal and her Multiplication Tables

Amal needs your help with her mathematical woes. Can you help her generate a multiplication table for a given number?

Constraints

- $1 \leq M, N \leq 100$

Interactions

The input comprises two space-separated integers denoting the values of N and M .

The output must contain a multiplication table for N with multiples from 1 to M inclusive..

Sample

Input	Output
2 6	2 x 1 = 2 2 x 2 = 4 2 x 3 = 6 2 x 4 = 8 2 x 5 = 10 2 x 6 = 12
11 3	11 x 1 = 11 11 x 2 = 22 11 x 3 = 33

In the first case, a multiplication table for $N = 2$ is generated for multiples from 1 to $M = 5$ inclusive.

In the first case, a multiplication table for $N = 11$ is generated for multiples from 1 to $M = 3$ inclusive.

Exercise

	Input
(i)	3 4
(ii)	4 3
(iii)	8 5

4. Stacking Squares

Shoukat wants to make a pattern of stacking squares, where each subsequent square is one unit of length larger than the previous one. Each square is made up of the digit that represents its size. Can you help Shoukat generate the pattern?

Constraints

- $1 \leq S \leq 9$

Interactions

The input comprises one integer denoting the value of S , the size of the largest and bottom-most square in the pattern.

The output must display all the squares from smallest to largest, each made up of a single digit representing its size.

Sample

Input	Output
2	1 22 22
4	1 22 22 333 333 333 4444 4444 4444 4444

In the first case, $S = 2$. The first square has a size of 1, which is made up of the digit 1. The second square has a size of 2, and occupies two rows. It is made up of the digit 2.

In the second case, $S = 4$, This draws the pattern from the previous example, then adds a square of size 3 made up of the digit 3 and occupying three rows, and then a square of size 4.

Exercise

	Input
(i)	1
(ii)	3
(iii)	5

For the following question, (a) dry run the proposed solution using the two inputs provided in the Sample section, (b) identify the error, and (c) provide a correct solution.

5. Strong Password

You are giving an interview for a cybersecurity internship. It has been going great, until the interviewer puts you on the spot and asks you to explain how you would code a program to ensure the user entered password is secure. He hands you a pen and paper and gives you 5 minutes to build a rough draft. Being overwhelmed, you are unable to come up with a good enough answer. The interviewer presents you with a solution. Unflagged, you decide to turn the tables on the interviewer and find weaknesses in his proposed solution.

The interviewer has written a function called *password_strength* that takes in a string parameter *password* and has a variety of possible return strings depending on the strength of the password. The interviewer has a set of criteria that the password must meet to be considered strong. These are:

- Length must be between 8 and 30 letters (exclusive).
- There must be at least one upper case character.
- There must be at least one lower case character.
- There must be at least one digit.
- There must be no spaces in the password.

If the first criterion is not met, the function returns “Length must be between 8 and 30 letters (exclusive).” If the first criteria is met and the second is not, then it returns “There must be at least one upper case character.” If the first and second criteria are met and the third isn’t, it returns “There must be at least one lower case character..” If first, second and third criteria are met but the fourth isn’t, it returns “There must be at least one digit.” If all but the fifth

criteria are met, it returns “There must be no spaces in the password.” If all criteria are met, it returns “Strong Password!”

Constraints

- *password* contains no special characters

Interactions

The input comprises a single line containing the string *password*.

The output must be a string as returned by the function *password_strength* when *password* is passed to it.

Sample

Input	Output
sd2j42n35	There must be at least one upper case character.
AsO3qsas	Length must be between 8 and 30 letters (exclusive)

In the first case, the password passes the first criterion but fails the second.

In the second case, the password fails the first criterion.

Proposed Solution

```
1 def password_strength(p):
2     if 8 <= len(password) and len(password) <= 30:
3         return "Length must be between 8 and 30
4         letters (exclusive)."
5         found_upper = False
6         found_lower = False
7         found_digit = False
8         found_space = False
9         for letter in password[8:30]:
10             if letter.isupper():
11                 found_upper = True
12             elif letter.islower():
13                 found_lower = True
14             elif letter.isdigit():
15                 found_digit = True
16             elif letter == ' ':
17                 found_space = True
18         if not found_upper:
19             return "There must be at least one upper case
20             character."
21         elif not found_lower:
22             return "There must be at least one lower case
23             character."
24         elif not found_digit:
25             return "There must be at least one digit."
26         elif not found_space:
27             return "There must be no spaces in the
28             password."
29         return "Strong password!"
```